1. MAINTENANCE REFERENCE DATA

- Equipment Model: HP-200 Forming Press (Machine ID: M-HP200).
- Hydraulic fluid specification: H46 hydraulic oil.
- Return-filter identifier: RF-20.
- Reservoir capacity & level: Maintain between MIN and MAX markings on sight-glass.
- Planned oil replacement interval: Every 2,000 operating hours or 12 months, whichever comes first.

2. INSPECTION BOUNDARIES & SAFETY PRECAUTIONS

Physical maintenance tasks must only be performed by authorized maintenance personnel.
All work must strictly adhere to facility isolation and stored-energy control procedures.
Safety Notice: The manual does not provide site isolation procedures. Personnel must never open pressurized hydraulic lines, remove protective safety guards, or alter internal machine valve settings.

3. RECORDING COMPLETED MAINTENANCE WORK

Whenever maintenance or inspection is completed, the technician must record:

- Machine ID (M-HP200).
- Current operating-hour reading.
- Date of inspection/service.
- Specific inspection item or component serviced.
- Physical observation and measured parameters.
- Action reference code.
- Authorized reviewer signature.

Important: An AI-generated suggestion or diagnostic note does not constitute completed maintenance.

1. SCHEDULED INSPECTION INTERVALS
- Before each shift: Inspect reservoir level and check for visible external leakage via sight-glass. Record oil level and leakage findings.
- Every 50 operating hours: Inspect the RF-20 return-filter restriction indicator. Record indicator status.
- Every 250 operating hours: Inspect cooling-air path and external hydraulic hoses for obstructions, physical damage, or leakage.
- Every 500 operating hours: Planned replacement of the RF-20 return filter. (Note: Earlier replacement is required if filter restriction is confirmed).
- Every 2,000 operating hours or 12 months: Planned replacement of H46 hydraulic oil (whichever limit is reached first).

2. RETURN FILTER (RF-20) & ALARM E42 REFERENCE
- E42 indicates restriction in the RF-20 hydraulic return filter on HP-200.
- Maintenance technicians must review the restriction indicator and previous service records.
- If restriction is confirmed, execute an authorized filter replacement.
- Note: The maintenance manual does not specify a calendar-based replacement interval for the RF-20 filter (only the 500 operating hours interval or condition-based replacement).

1. LOW HYDRAULIC PRESSURE (E17) TROUBLESHOOTING OBSERVATIONS
When investigating alarm E17 on the HP-200, collect observations in the following mandatory order:
1. Reservoir level: Verify whether hydraulic fluid is between MIN and MAX markings.
2. Visible leakage: Inspect fittings, hoses, and manifolds for external leakage.
3. RF-20 restriction indicator: Verify whether the return-filter restriction indicator is tripped.
4. Oil temperature and recorded pressure during the forming stage (compare against normal 120-140 bar).
Important Diagnostic Note: Low oil level, leakage, or return-filter restriction are diagnostic observations; they do not individually prove the root cause. The manuals do not provide a diagnostic test that conclusively proves hydraulic pump failure. If E17 low pressure repeats, keep machine on HOLD and escalate.

2. HIGH OIL TEMPERATURE (E24) TROUBLESHOOTING
When investigating alarm E24, inspect:
- Cooling-air path: Check for dust accumulation, physical debris, or airflow obstruction.
- Cooling-fan status indication.
- Reservoir oil level.
- Recent machine operating cycle patterns.
Note: The manual does not provide cooling-fan replacement instructions or guaranteed cooldown times.

3. ESCALATION RECORD REQUIREMENTS
Escalation reports to authorized engineering must contain:
Machine ID, timestamp, operating hours, alarm code, pressure, oil temperature, reservoir level, leakage observations, RF-20 filter indicator state, cooling-fan indication, prior incident reference ID, and reviewing technician.

4. RETURN-TO-SERVICE AUTHORIZATION
Return-to-service can ONLY be authorized by the designated supervisor following technical maintenance review. The AI assistant cannot authorize machine restart, nor does alarm clearance alone permit restarting.